Chapter 13
Potato Leaf Disease Detection and Classification Using VGG16

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1 Introduction

Potato (Solanum Tuberosum) is an essential and widely cultivated crop that belongs to the Solanaceae family. It is a starchy tuberous crop that serves as a staple food for millions of people around the world. Because India's economy relies mainly on agriculture with a diverse crop portfolio, potatoes are a significant part of their society. India is now the second leading producer of potatoes globally, producing over 43 million tons in 2018. Nonetheless, despite the increasing global demand and the opportunity for India to export more, potato production expansion arises as the most important factor. Unfortunately, this is not easy, as significant diseases such as early blight and late blight have caused potato exports and yields to decline in recent years. As a result, local farmers are facing various challenges. Early and late blights are common potato diseases that affect farmers in the region. The symptoms of late blight disease can be observed as blistered spots on leaves that eventually decay and dry up, while early blight disease manifests as small, black lesions. The use of convolutional neural networks (CNN) and the VGG16 algorithm will be of great benefit to farmers,

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as it can help distinguish between these diseases and potato leaves. There are three categories of processed images available, which include healthy leaves, early stage late blight, and fully developed late blight. The pictures are divided into two groups, one for training and the other for validation, to help farmers identify and treat these diseases using appropriate fertilizers.

Recently, there has been a growing interest in using advanced technologies such as computer vision and deep learning in identifying and classifying plant leaf diseases. Convolutional neural networks (CNNs) have shown great potential in tasks such as detecting and diagnosing plant diseases using image identification. Several studies have been conducted to explore new perspectives on the characterization of plant diseases using deep learning techniques like CNNs, fuzzy image enhancement methods, and VGG16 algorithm. These approaches aim to create accurate and reliable detection methods for plant diseases using computer-assisted technologies. In this context, developing a CNN-based system for potato leaf disease detection and classification holds immense potential to improve disease management and increase crop productivity. By automating this process, farmers and agricultural experts can quickly identify diseased plants, implement targeted interventions, and minimize crop losses.

2 Literature Review

Dataset: The model is trained using a dataset, which is a collection of data. Image information: The project's dataset is the Kaggle PlantVillage dataset. Deep learning encompasses a variety of algorithms, including convolutional neural networks (CNNs), which are commonly used for tasks such as image classification and object recognition. CNNs are particularly suited for analyzing visual data using artificial neural networks, and they employ a grid-like structure that allows them to process data in a more efficient manner. Through the use of CNNs, deep learning has shown promise in a range of applications, from medical image analysis to computer vision and object detection tasks, despite the fact that there are various types of neural networks.

The author has detected the diseased leaf and area of the leaf affected using CNN model and has achieved an accuracy of 95.86% [1]. The author has presented a system that uses deep learning to classify two types of diseases in potato plants based on leaf conditions, using the GoogleNet, ResNet50, and the VGG16 convolutional neural network architecture model to create an accurate classification system. The model has achieved 97% accuracy for the first 40 CNN epochs, indicating the feasibility of the deep neural network approach [2]. In another analysis, the author has performed picture division with more than 2000 pictures of healthy and unhealthy potato's leaf, which are collected from Kaggle and a few preprepared models are utilized for acknowledgment and classification of healthy and diseased leaves. The proposed model has predicted with an accuracy of 91.41% in testing with 30% test data and

70% train data [3]. A novel plant leaf disease classification with image processing technique is introduced, and best results were obtained with very little computational effort [4].

The other approach proposed has used CNN with a maximum of 50 epochs for the training model, with a batch size of 32. The model was observed to outperform with 99.91% accuracy during training, 99.80% accuracy during validation, and 100% accuracy in the testing phase [5]. The CNN model was used and deployed on the Streamlit platform to classify the leaf disease [6]. In another research, the author has modified the existing MobileNet-V2 architecture using the fine-tuning approach and the proposed approach separately attained an average accuracy and specificity of 97.33 and 98.39% on the locally collected image dataset [7]. In the papers [8, 9], the author has performed feature extraction with *K*-means clustering and classification with ANN and has outperformed the existing models.

A new work is done in detecting plant diseases using OpenCV in [10]. In [11], the author has selected VGG16, VGG19, and ResNet50 network models. Using VGG16, the author has achieved 97% accuracy which is the best-provided results among three networks. In [12], the author has performed disease classification with image processing and random forest technique and the average accuracy achieved was 93%. In [13], the author has performed the detection of plant diseases by analyzing the texture of leaf using ANN classifier. In [14], the author has presented a system to classify the four types of diseases in potato plants based on leaf conditions by using VGG16 and VGG19 convolutional neural network architecture models to obtain an accurate classification system. This experiment has achieved an average accuracy of 91%. In [15], the author has used the pretrained model VGG 19 to predict the diseases in potato crops using leaf images and has achieved an accuracy of 97.8%.

3 Proposed System

In this work, VGG16 is used to distinguish between diseased and unimpaired leaves as well as to spot defects in affected plant leaves. Using pictures to train the model and the input leaf to decide output, the CNN model is made to work with both healthy and diseased leaves. In the same way that human eyes and brains do, VGG16 can identify as many images as it can at a time. We are incredibly adept at observing, comprehending, picking up on, and inferring from a scene because it occurs in our brains so quickly. CNNs are necessary to help a computer comprehend and interpret a scene. The proposed work will be able to detect the diseased area on the leaf and classify diseases into late blight, early blight, or a healthy leaf.

The architecture used is VGG16, and the dataset used is the PlantVillage dataset, which consists of numerous images of different leaves. The dataset is then trained using VGG16. The image needs to be processed in order to find the sickness on the leaves. Image processing is an important field that has many practical applications, including in computer vision and medical imaging. The process begins with image acquisition, which is also known as pretreatment. Once the image is acquired, a

variety of techniques can be used to extract specific features or enhance its appearance, including image enhancement and restoration. Difficult characteristic of image processing is segmentation, which involves splitting an image into its discrete objects or components. Despite its challenges, image processing technology continues to advance, with new research focusing on areas such as artificial intelligence and machine learning. These are the steps involved in image processing. Then the final model is built and executed.

3.1 Architecture

This architecture in Fig. 1 illustrates that it gathers a dataset of potato leaf images, including both healthy leaves and different diseased leaves. Preprocess the images by resizing them to a constant size, then split the dataset into two different sets namely training set and validation set. Transfer learning mechanism helps to improve the performance and its working mechanism is shown in Fig. 2. Here, the VGG16 architecture has feature extraction by removing the final fully connected layers and the adding custom layers for the disease classification, and the working model of VGG 16 is shown in Fig. 3. Then it tests the trained model on the validation set to access its performance metrics and its generalization ability.

3.2 Transfer Learning

Transfer learning is a machine learning technique, learning new task is dependent on prior knowledge.

3.3 VGG 16 Architecture

The VGG16 architecture is a convolutional neural network architecture that was developed by Visual Geometry Group and its input to the VGG16 network is an image of fixed size and it contains convolutional blocks which consist of five groups of convolutional layers, each followed by a max pooling layer. After the convolutional and pooling layers, the network ends with fully connected layers. For the disease detection here, we remove the last fully connected layers and replace them with custom classification layers.

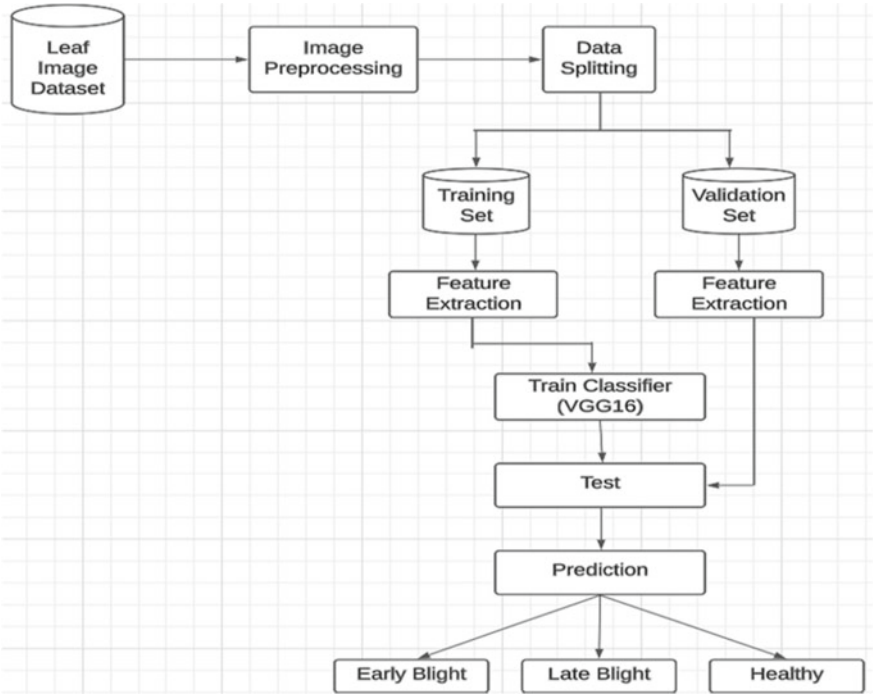


Fig. 1 Architecture

4 Modules

4.1 Image Processing

Image preprocessing is the practice of performing image processing on digital images using computer algorithms. By using a specialized algorithm to analyze the image, we can identify the plant. With a particular algorithm, we apply a similar strategy for picture processing and detection. In this process, the image quality is crucial since without a clear image, the algorithm would not work.

4.2 Data Splitting

Data splitting is a method of separating data into two or more parts. Testing or evaluating the data in one part and training the model in the another portion of a two-part split is common. It helps to guarantee the precision of procedures that use data models, such as machine learning. A new approach can be used to effectively separate data for the identification of plant leaf diseases. First, a dataset with pictures of both

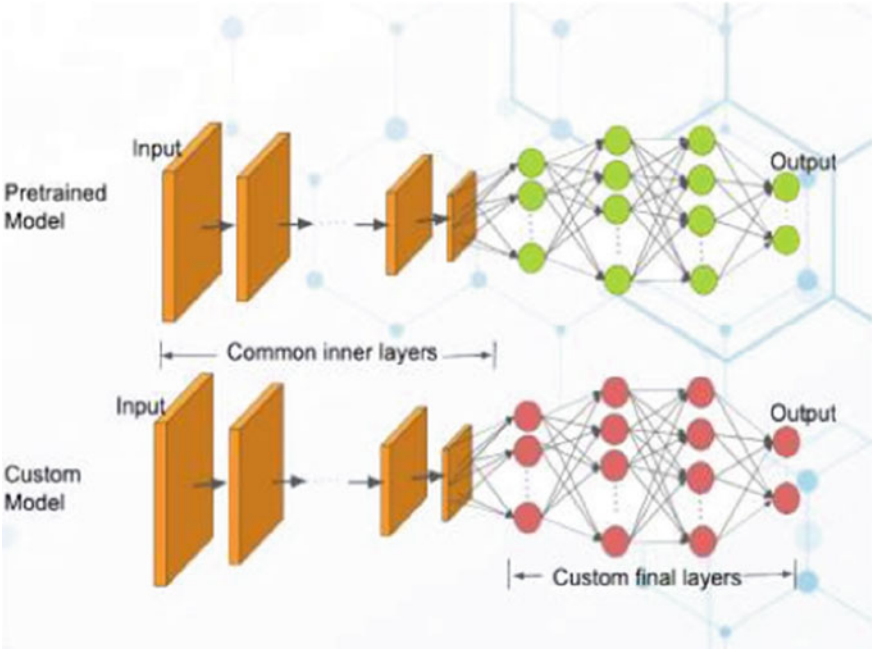


Fig. 2 Transfer learning mechanism

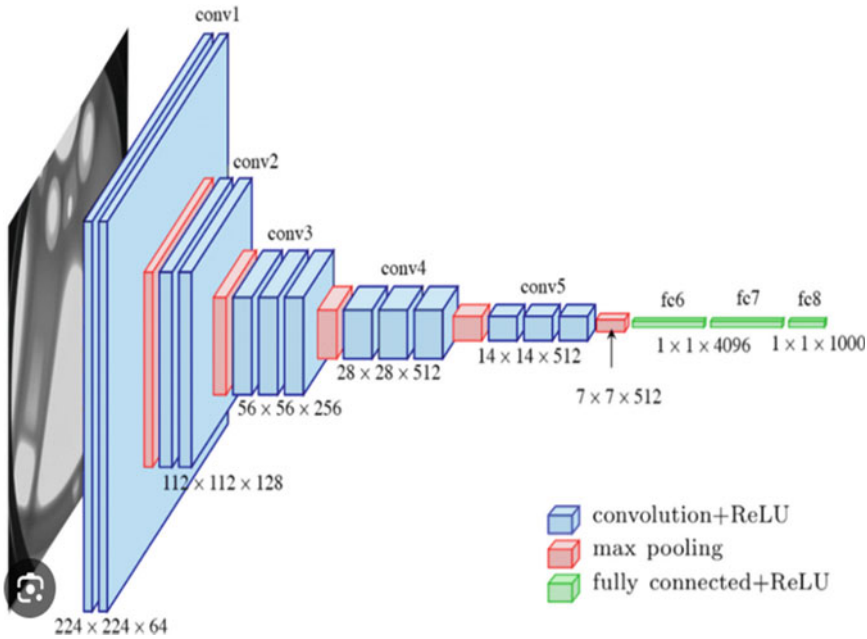


Fig. 3 VGG 16 architecture

healthy and sick plant leaves is acquired. A wide and representative collection of photos that includes various plant types, illnesses, and environmental circumstances must be developed. Training, validation, and testing are the three main subsets that can be extracted from the dataset. The training set makes up the majority of the data, often between 70 and 80%. The model is trained using it to discover the fundamental patterns of both healthy and sick leaves.

4.3 Feature Extraction

Feature extraction is a fundamental step in deep learning that involves identifying and extracting relevant patterns or features from raw data. These extracted features represent the most important aspects of the data and are used as input for subsequent layers in a deep neural network. Through a sequence of convolutional and pooling layers, CNNs are specially made to capture the spatial and hierarchical information contained in images. The qualities of the leaf are fully represented by integrating these various elements. With the use of this feature set, machine learning or pattern recognition algorithms may accurately classify and identify leaf diseases. Effective crop management and disease control techniques are aided by automated systems' quick and accurate ability to recognize and diagnose plant diseases.

An overview of how feature extraction functions in CNNs is given below:

Convolutional Layers: Convolutional filters are used in the first layers of a CNN to scan the input image in a sliding window fashion. Each filter creates a feature map by performing element-wise multiplication on a tiny portion of the input image using its weights. At various spatial scales, these filters record a variety of visual patterns, including edges, textures, and forms. CNNs may simultaneously extract several different kinds of information by using several filters.

Pooling Layers: By condensing and lowering the spatial dimensions of the feature maps, pooling layers down sample the data. Max pooling, which selects the highest value within a narrow window as the typical value for that area, is a common technique. By establishing translation invariance by pooling, the CNN becomes more resistant to minute changes or variances in the input.

Fully Connected Layers: These are used in categorizing the learned characteristics in the final stages of a CNN. These layers link every neuron from the prior layer to the following layer, enabling the network to understand how the extracted features relate to the intended job, such as classifying or identifying particular objects.

4.4 Data Classification

Data classification is essential for identifying plant leaf diseases. The next stage is to categorize the leaf images into various disease groups after the preprocessing methods have been applied. Each image of a leaf is classified according to the severity of the disease present in that leaf.

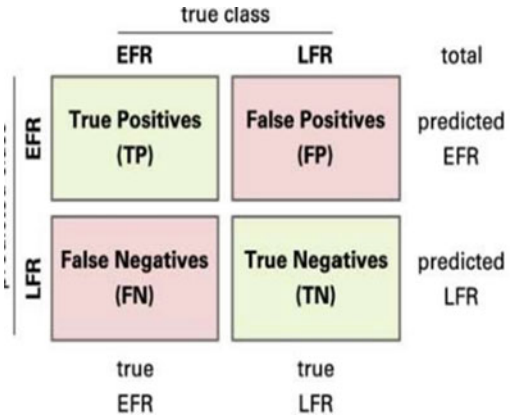
For this objective, a variety of classification techniques, from basic machine learning algorithms to deep learning models, can be used. These algorithms create a predictive model using the preprocessed leaf images and the related disease diagnoses. The data patterns and attributes that distinguish healthy leaves from diseased ones and distinguish between various types of diseases are captured by the model.

In conventional machine learning methods, features are often extracted utilizing methods from the preprocessed images. Convolutional neural networks (CNNs) are an acclaimed type of deep learning used for various applications, including the detection and classification of plant diseases by analyzing leaf health. In a CNN, the layers are grouped into different functional categories, with one or more convolutional layers followed by subsampling layers and fully connected layers. Each feature layer gets a small input feature set from the previous layer. CNNs have been utilized with great success in identifying and classifying the various plant leaf diseases, which is crucial for monitoring plant health in agriculture.

4.5 Testing

Testing is the procedure of assessing the efficiency and precision of a trained CNN model on unobserved data. It is a critical stage in the creation and application of CNNs since it enables us to evaluate how well the model generalizes to novel, unexplored data and measures its efficiency in completing the task at hand, such as segmentation, object recognition, or picture classification. In the testing phase, the trained CNN model is applied to a different dataset known as the test set, which contains instances not utilized in training or validation. This division is sure that the performance of the model is estimated using data that it has never seen before, resulting in an objective assessment. The confusion matrix shown in Fig. 4 helps us in finding the accuracy of the proposed model with the following four parameters.

True Positives: Expected correct and output also correct.
True Negative: Expected wrong and output also wrong.
False Positive: Expected correct but the output was wrong.
False Negatives: Expected wrong but correct.



$$PR = \frac{TP}{TP+FP}$$

$$RE = \frac{TP}{TP+FN}$$

$$CA = \frac{TP+TN}{TP+TN+FP+FN}$$

$$F_1 = \frac{2TP}{2TP+FP+FN}$$

Fig. 4 Confusion matrix for a dataset

5 Results

In the proposed effort, experiments make use of the PlantVillage dataset. Work is done on potato plant species. The technique can identify three different kinds of diseases of a potato leaf: early blight, late blight, and healthy leaves. The VGG16 model has shown effectiveness in the diagnosis of plant leaf diseases. A DCNN architecture called VGG16 is renowned for excelling at image categorization tasks. The VGG16 model can learn detailed patterns and features linked to various types of illnesses by training it on a large number of identified leaf photos. The VGG16 model can properly categorize and identify the presence of disease with a high degree of accuracy when given new leaf images. Early identification and prompt response are made possible by this, which results in efficient disease management techniques and increased crop health. Utilizing VGG16 to identify plant leaf diseases demonstrates how deep learning models have the ability to revolutionize agricultural methods and guarantee sustainable food production. The predictions of VGG 16 model are shown in Figs. 5 and 6. Training and validation accuracy, training and validation loss of the model are shown in Fig. 7.

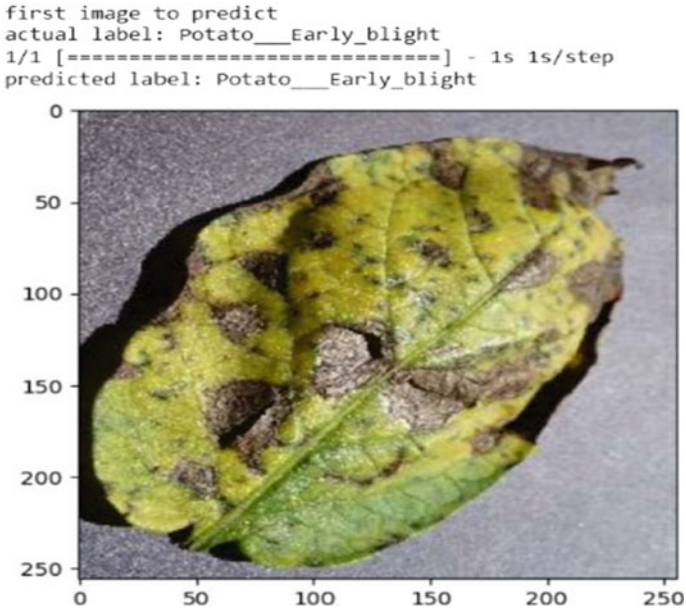
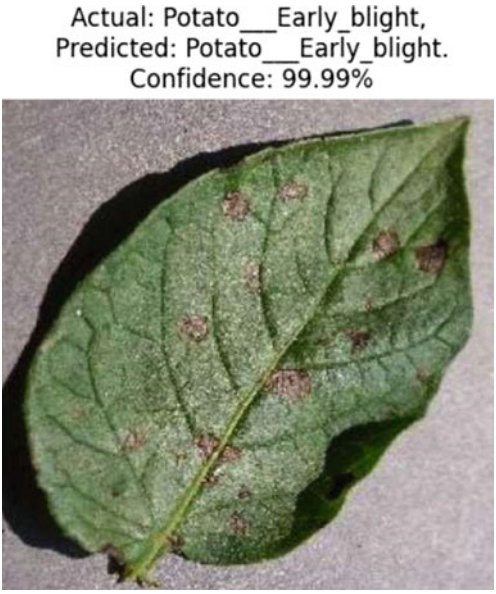


Fig. 5 First image to predict

Fig. 6 Potato early blight leaf



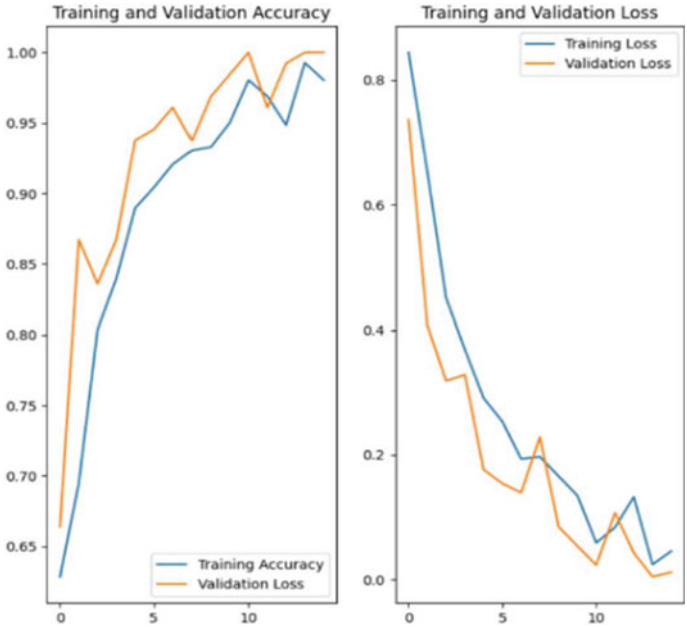


Fig. 7 Training and validation accuracy and training and validation loss

6 Conclusion

The suggested model uses convolutional neural networks to create a plant disease detection system. As a classifier and feature extractor, VGG 16 is employed. On different varieties of potato plants, experiments are done. By adding more data and experimenting with various optimizers, deep learning-based architecture delivers notable results that can be further enhanced.

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